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DAT321 DIT843 Sustainability and Software Quality

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Alexander Serebrenik

- Professor at the Eindhoven University of Technology
- Major contributions to diversity and inclusion in SE.
- Inaugural lecture on Social SE: https://youtu.be/69CRnH3IbDE
- LGBTQIA+ representation in Software Engineering academia.

What we will learn today?

- What is **Sustainability**? How can we use it as a quality attribute?
- What are sustainability **impacts**? How is it related to IT?
- What are immediate, enabling and structural **effects**?
- What are the **five dimensions** of sustainability?
- What is the **Sustainability Awareness Framework**? How can we use it?

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Part 1:

Sustainability

And IT

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Sustainable Development

- United Nations - Sustainable development:

> "development which meets the **needs of the present without compromising** the ability of **future generations** to meet their **own needs**"

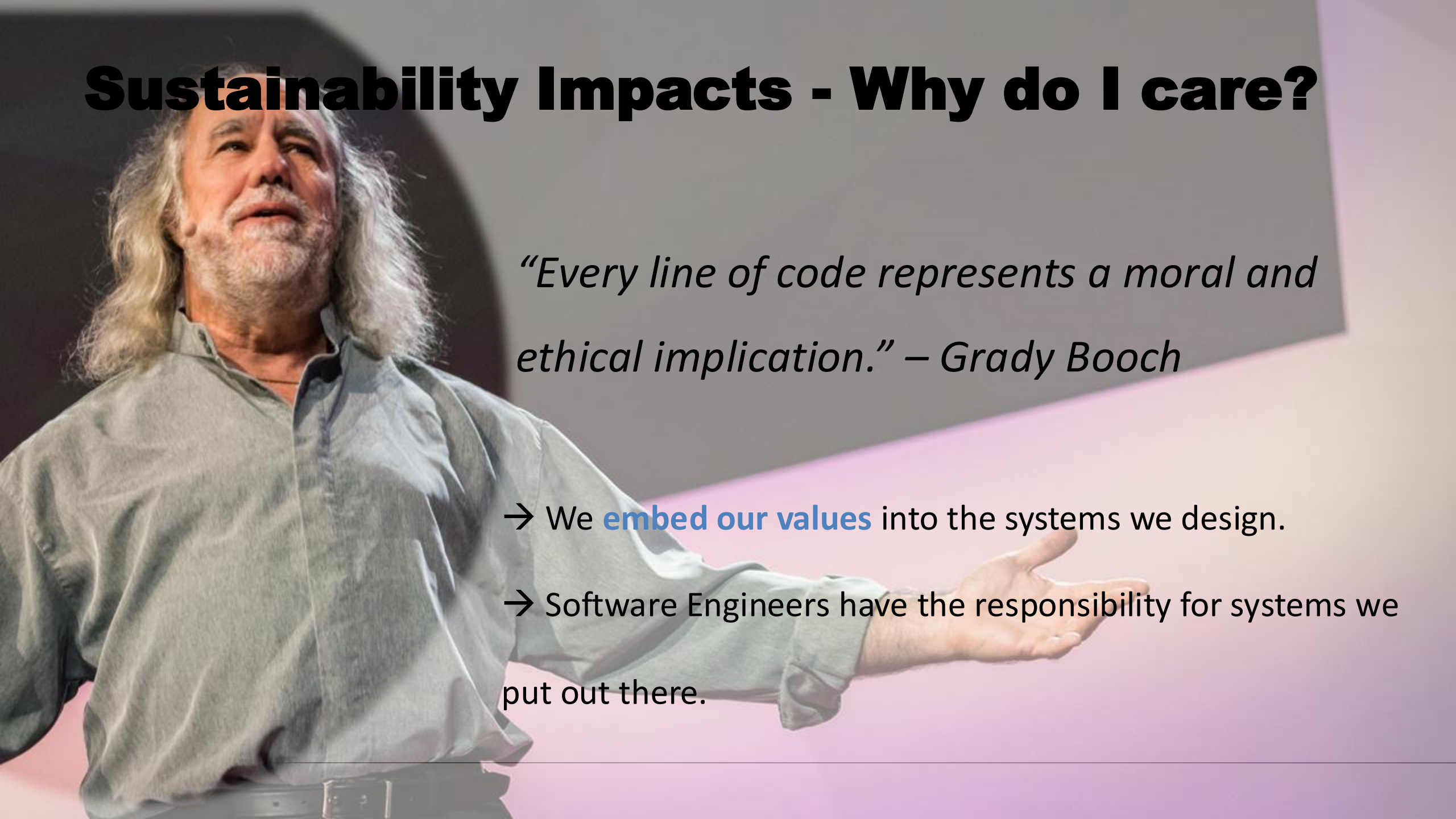
- Use of Sustainable Development Goals (**SDG**)

http://www.un.org/wcm/webdav/site/climatechange/shared/gsp/docs/GSP1-6_Background%20on%20Sustainable%20Devt.pdf

UN Sustainable Development Goals

https://sustainabledevelopment.un.org/focussdgs.html

Sustainability Impacts - Why do I care?



"Every line of code represents a moral and ethical implication." – Grady Booch

→ We **embed our values** into the systems we design.

→ Software Engineers have the responsibility for systems we put out there.

What is sustainable SE?

- **Defining and developing** software products in a way so that negative and positive **impacts on sustainability** (resulting from the SW product) are **continuously** assessed documented and optimized.

- Impact means that the software will **affect** energy, resource demand, economic development and the daily life of individuals and communities.

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Example – Airbnb - Envisioned benefits

MARKETPLACE TO RENT YOUR HOUSE

MORE AVAILABILITY OF CHEAPER, SHORT-TERM RENTAL OPTIONS

LOCAL ACCOMMODATION

Things **foreseen at design time**. Anything missing?

Example – Airbnb – Gone wrong

Economic and social impacts.

NEW YORK: HOMEOWNERS CAN EARN *55%* MORE THAN THE MEDIAN LONG-TERM RENTAL

ESTIMATED: AIRBNB *REMOVED* 7,000 - 13,000 UNITS OF HOUSING IN NY → *INCREASE OF 1.4% IN THE MEDIAN LONG-TERM RENT*

72% OF POPULATION IN NEIGHBOURHOODS AT HIGHEST RISK OF AIRBNB-INDUCED *GENTRIFICATION* ARE NON-WHITE → *INCREASING RACE SEPARATION*

D.Wachsmuth and A. Weisler. Airbnb and the rent gap: Gentrification through the sharing economy. *Environment and Planning A: Economy and Space*, 50(6):1147–1170, 2018.

Anything else?

Have you experienced negative impacts form technology?

- Making sense of the world: (Misinformation, fake news)
- Attention and cognition: (Loss of crucial abilities)
- Physical and mental health: (Stress, loneliness, addiction)
- Social relationships: (Less empathy, more confusion)
- Politics and elections: (Propaganda, distortion)
- Systemic oppression: (Amplification of sexism, racism)
- The next generations: (Developmental delays, suicide)

Center for Humane Technology

https://ledger.humanetech.com

Part 1: Take-away

- We are the ones designing and deploying software to end-users, hence **it is our responsibility** to deliver software that **meet sustainable goals**.

- We need to be **aware of the different ways** in which our software can impact users and society, particular to **introduce systemic problems**.

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Part 2:

Sustainability as a Quality Attribute

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Sustainability and the Quality Attributes

- Sustinability was **not one of the 8 original quality attributes** proposed by the ISO 25010:2011 standard.
- The most important aspect of using sustainability as a quality attribute is to **bring awareness** to different types of impact.

- We use **two aspects** to describe sustainability in SE:
  - Sustinability dimensions, and Sustainability impact.

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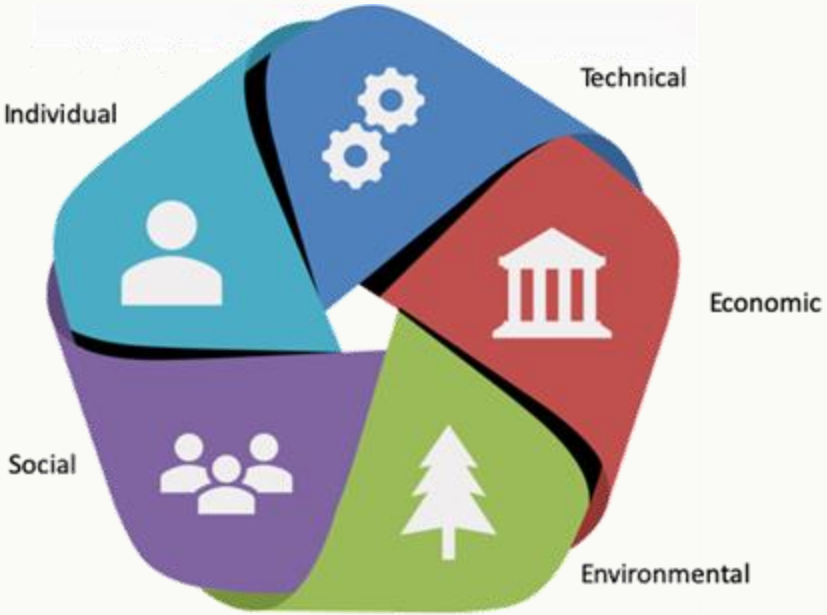
Sustainability Dimensions

- We work with Five Dimensions where we want to foster sustainability in connection to software development activities:

Impact on human resources on **personal level**.

Impact on **communities**



Impact on **use and evolution** of **technology**

Impact on **capital and value** generated by our applications

Impact on **natural resources**

B Penzenstadler, H Femmer. A generic model for sustainability with process-and product-specific instances. Proceedings of the 2013 workshop on Green in/by software engineering, 3-8

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Hilty, and Aebischer. "ICT for sustainability: An emerging research field". ICT Innovations for Sustainability. 2015

What are sustainability impacts?

	IT as part of the problem	IT as part of the solution	
Technology	Life-cycle of IT: Production, Use, Disposal		**1** Immediate effects
enables			
Application	Induction effects; Obsolescence effects	Substitution effects; Optimization effects	**2** Enabling effects
enables			
Behavioral and structural change	Rebound effects; Emerging risks	Transition towards sustainable patterns of production and consumpt.	**3** Structural effects

Sustainability Awareness Framework

- Set of **templates**, questions and forms to be used during software development to **help practitioners to identify** the immediate, enabling and structural effects of their application on **sustinability goals**.
- Connected to each of the **5 dimensions**.
- Bring **awareness** to risks and impact that might have been missed.

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Sustainability Awareness Framework

Warm-up	Capture	Analysis	Synthesis
Moderator welcomes and explains logistic	Moderator leads through dimensions w. questions	Moderator presents and asks open questions	Moderator facilitates synthesis of chains of effects into diagram(s).
Company introduces product or system	Stakeholders brainstorm, reflect and discuss	Group classifies effects into three orders	Detail diagrams can be added for in-depth visual
Moderator introduces SusAF	Scribe takes notes and denotes cross references	Group identifies chains of effects	Threats & opportunities get summarised in report
		Stakeholders discuss relations across dim.	
		Potential actions to mitigate and change	

Duboc, L., Penzenstadler, B., Porras, J., Kocak, S. A., Betz, S., Chitchyan, R., ... & Venters, C. C. (2020). Requirements engineering for sustainability: an awareness framework for designing software systems for a better tomorrow. Requirements Engineering, 1-24.

https://zenodo.org/record/3676514#.YEIC7y1Q0_X

Exercise – Google Maps as a Case

- You are hired to work at **Google Maps** with the features below. Choose one feature and suggest a **sub-feature.** For that sub-feature, write the immediate, enabling and structural effect.
  - F1: Users can sign in to the platform to save favorite locations, travel history, and preferences.
  - F2: The system must provide accurate and continuously available map data and navigation services, even under high traffic conditions.
  - F3: Route calculations, map rendering, and real-time traffic updates must be delivered with low latency to handle large-scale geographic data.
  - F4: The interface should enable users to quickly search for locations, understand routes, and interact with the map.

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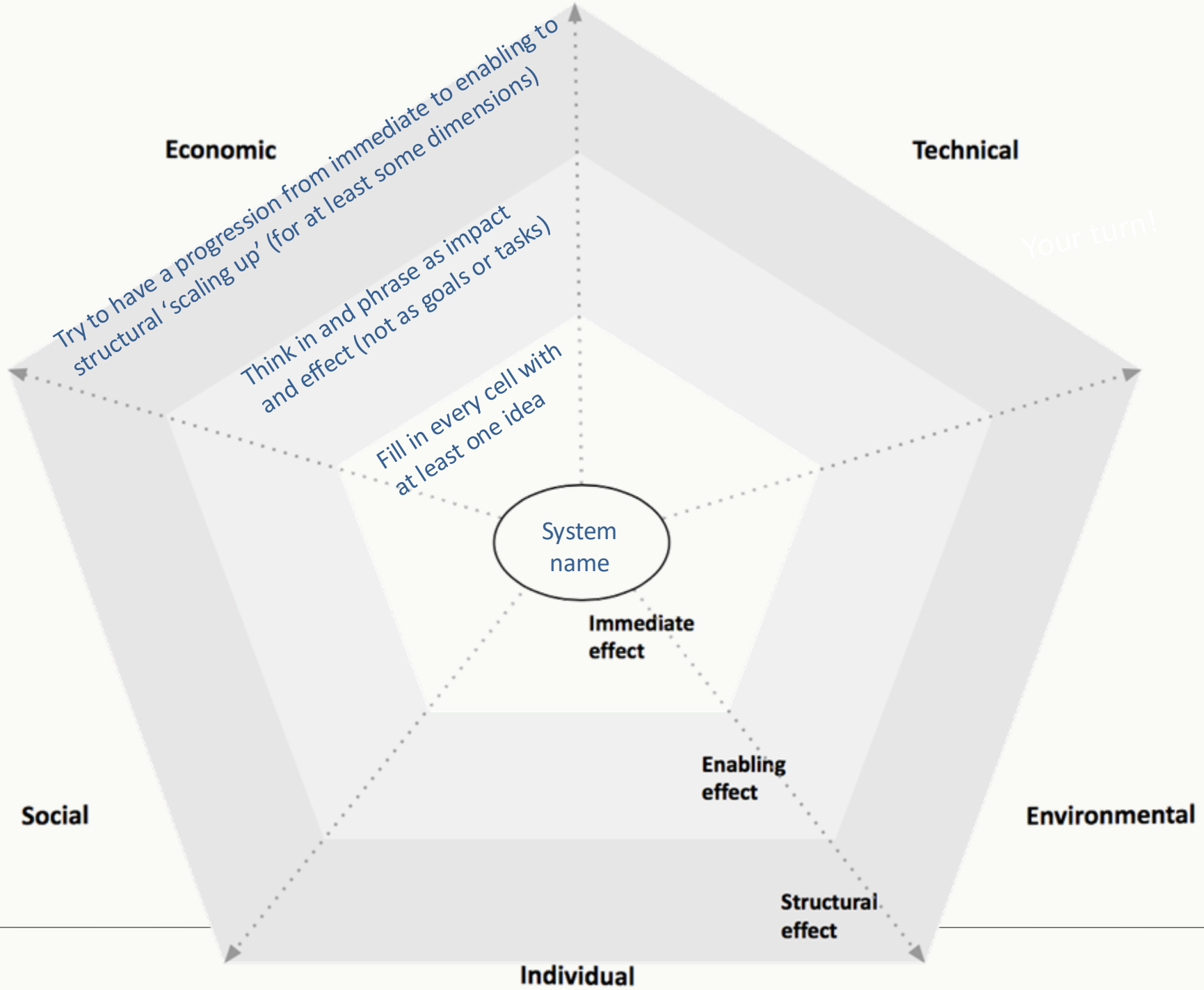
Economic
Technical
Social
Environmental
Individual
Try to have a progression from immediate to enabling to structural 'scaling up' (for at least some dimensions)
Think in and phrase as impact and effect (not as goals or tasks)
Fill in every cell with at least one idea
Your turn!
System name
Immediate effect
Enabling effect
Structural effect

Part 2: Take-Away Points

- Sustainability is **pervasive to all quality attributes** that we have seen.
- There are **five dimensions** to sustainability: Individual, Technical, Economical, Social and Environmental.
- Our software application have **immediate, enabling and structural** effects on sustainability.
- Recent work focuses on **bringing awareness** to the impact of our software applications to sustainability during all stages of development.

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